Feature Balance for Fine-Grained Object Classification in Aerial Images

Wenda Zhao, Tingting Tong, Libo Yao, Yu Liu, Congan Xu, You He, and Huchuan Lu, *Senior Member, IEEE*

***Abstract*— Fine-grained object classification (FGOC) focuses on identifying subcategories of objects, which is crucial in military and civilian. Existing FGOC methods primarily focus on high-resolution aerial images, limiting their application on low-resolution (LR) FGOC that is a more realistic setting, especially on resource-constrained satellite devices. It is more challenging to deal with LR FGOC since objects' details are blurred or missing. Addressing this issue, we make the first attempt to explore LR FGOC and propose a novel pipeline based on two technical insights: 1) feature balance strategy discriminatively integrates super-resolution weak and strong detailed presentations into coarse features of LR aerial images, achieving a feature balance to avoid that the weak detailed presentations are inhibited by the strong ones and 2) iterative interaction mechanism alternately refines feature details of the discriminative ship regions and optimizes the performance of FGOC. Moreover, we build a low-resolution fine-grained object (LFS) dataset to promote further study and evaluation. Extensive experiments on the proposed LFS dataset and the other three object datasets of DOTA, FS23, and HRSC2016 demonstrate that our method outperforms state-of-the-art algorithms. Dataset and code are publicly available at https://github.com/wdzhao123/FBNet.**

***Index Terms*— Aerial images, feature balance, fine-grained object classification, iterative interaction, low resolution.**

I. Introduction

FINE-GRAINED object classification (FGOC) aims at identifying subcategories of objects, possessing many essential applications, e.g., ship classification for maritime traffic, fishery management, and maritime military security [1]. Recently, deep learning has become a powerful means of studying fine-grained classification in aerial images, leading to remarkable breakthroughs [2]–[4]. However, existing methods often adopt high-resolution (HR) aerial images to obtain high performance, limiting their application on low-resolution (LR) aerial images, which is a more realistic setting, especially on resource-constrained satellite devices.

Current fine-grained classification methods are usually explored from three perspectives.

1) Focusing on discriminative regions through attention strategy [5]–[7], where the discriminative regions are selected to boost classification, such as progressive attention networks to learn rich part hierarchies [6] and mask-CNN model to localize parts and select descriptors [7].
2) Implementing local and global representation integration [8], where multilevel features are combined to improve fine-grained recognition, e.g., bidirectional long short-term memory network to fuse and encode the partial information of the complementary parts into a comprehensive feature for image classification [9].
3) Designing specific loss function [10], such as gradient-boosting loss that focuses on the confusing classes to move swiftly along the direction on the loss surface to resolve ambiguities [11] and mutual-channel loss that forces all feature channels of the same class to be discriminative [12]. Nevertheless, these methods generally use HR images as the inputs, and their performance degrades when dealing with the LR aerial images, as shown in Fig. 1(a).

We propose to explore LR FGOC, which is a more realistic application yet much less studied in the field of remote sensing. It is more challenging since the extremely important clues (detailed presentations) for FGOC are blurred or missing in LR aerial images. We propose a novel pipeline based on two technical insights.

1) *Feature Balance Strategy:* It discriminatively treats super-resolution weak and strong detailed presentations when they are combined to coarse features[1] of LR aerial images, avoiding that the weak detailed presentations are inhibited by the strong ones.
2) *Iterative Interaction Mechanism:* It alternately refines feature details of the discriminative object regions and improves the performance of FGOC.

Image super-resolution (SR) [13] focuses on reconstructing detailed presentations from the low-resolution input, and the detailed presentations can be integrated to LR features for improving the performance of FGOC. Existing feature integration strategies (e.g., sum, concat, attention weighting, direct

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[1]The features of object details in LR images are blurred, which are defined as coarse features.

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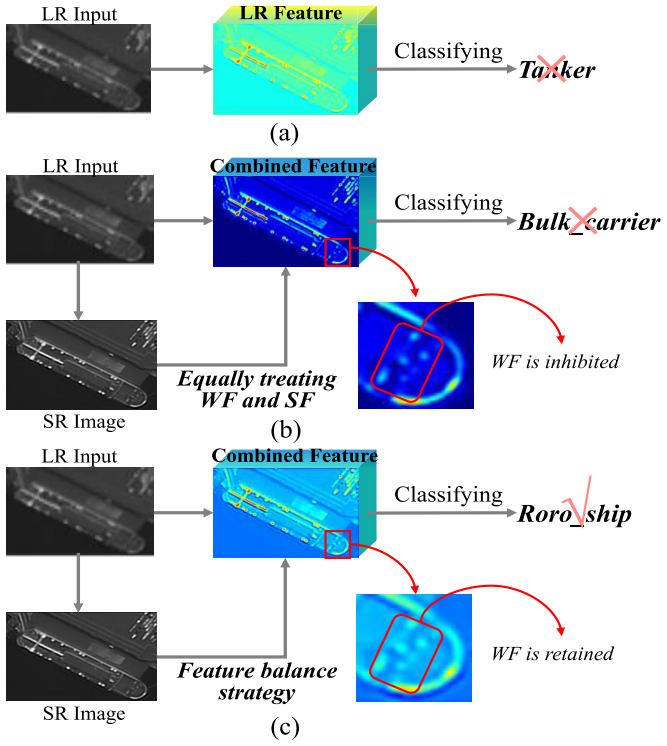


Fig. 1. Motivation of the proposed feature balance strategy. (a) Current fine-grained classification pipeline. (b) SR-based feature integration strategies equally treat SR's WFs and SFs. (c) Feature balance strategy. Even though the detailed presentations are combined to coarse features of LR image, the ship is erroneously classified since WFs are inhibited by the relatively SFs. In contrast, the proposed feature balance strategy efficiently combines WFs to LR features, helping the classifier make a proper decision.

supervision [14], and relationship transfer [15]) equally treat the weak and strong features of SR when they are furnished to LR features, where the weak features will be restrained by strong features [see Fig. 1(b)]. Instead, we schedule the feature integration according to their intensity consistency measure, i.e., if the features in LR image are weak or missing, detailed presentations in SR image are more integrated. Conversely, if the features in the LR image are strong, detailed presentations in the SR image are less integrated. Therefore, a feature balance in the LR image is achieved, guaranteeing that the weak features of SR are efficiently integrated without being inhibited by strong features [see Fig. 1(c)].

A model trained with classification labels tends to discover the most discriminative ship regions, which can induce the super-resolution network to pay more attention to reconstruct detailed presentations of these regions. Therefore, we utilize the class activation map [16] generated in the previous object classification network to structure an induced super-resolution loss, refining feature details of the corresponding discriminative object regions. Then, the boosted detailed presentations are adopted again to optimize the feature balance, improving the accuracy of FGOC. The interaction is iterative several times to obtain the final FGOC.

To achieve the study of FGOC in aerial images, datasets are also crucial. Existing optical aerial datasets are mainly constrained by two aspects. On one hand, there are few object subcategories, e.g., MASATI [17] includes ships and nonships and BCCT200 [18] consists of cargo ships, container ships, barges, and tankers. On the other hand, the objects' scales are usually large, e.g., high-resolution optical aerial images in FS23 [2] and HRSC2016 [19]. Addressing these limitations, we build a low-resolution fine-grained object dataset, named LFS. LFS contains military ships and civil ships, covering 12 subcategories. The ships are mainly concentrated at small scales, e.g., the percentage of ships with widths' range of $4 \sim 24$ pixels is 56% in LFS. Please see Section IV for more details.

The primary contributions are summarized as follows.

1) We make an attempt to explore the low-resolution fine-grained object classification, which is a more realistic setting on resource-constrained satellite devices.
2) We propose two key technical insights: feature balance strategy for efficiently integrating detailed presentations from super-resolution network and coarse contents from the low-resolution network, and iterative interaction mechanism for gradually optimizing the performance of FGOC.
3) We construct a low-resolution fine-grained object dataset, LFS, containing more ship subcategories with small scales, and implement the explicit training–testing protocol. Comprehensive experiments on four datasets of LFS, DOTA, FS23, and HRSC2016 are conducted to evaluate and verify the superior performance of the proposed feature balance framework. Our dataset and code will be publicly available to promote research in this field.

The remainder is organized as follows. Related work and the differences of our method are discussed in Section II. Then, our method is introduced in Section III. After that, we introduce the proposed datasets in Section IV. Section V builds extensive experiments to analyze and verify our framework. Finally, Section VI summarizes our work.

II. Related Work

Several tasks are related to our work: fine-grained object classification, image super-resolution, and auxiliary knowledge-based tasks. We review them briefly as follows.

A. Fine-Grained Object Classification

Learning and utilizing the most discriminative parts in an object is prevailing in the fine-grained image classification [6], [20]–[22]. For example, Ji *et al.* [5] and Liu *et al.* [23] designed attention mechanisms to obtain discriminative regions, and Ge *et al.* [9] and Wang *et al.* [24] implemented the detection technology to locate discriminative parts. Besides, high-order feature interactions [6], [25] and specific loss functions [10], [11] are explored to improve the performance of fine-grained object classification.

Recently, great progress has been made for the object detection and classification in the field of remote sensing [26]–[31]. Attention mechanisms are mainly studied and utilized. For instance, Sumbul *et al.* [3] classified the objects

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through using attention-driven multisource deep feature representations. Zhang *et al.* [2] introduced an RNN-based attention mechanism to obtain discriminative feature representations as the basis of the ship classification. Multiscale discriminative regions are localized by attention networks and then are used to scene classification in [32].

Existing fine-grained object classification methods generally utilize high-resolution images as the inputs to obtain good performance, and their applications are limited, especially on aerial images with lots of small-scale objects. In contrast, we explore the low-resolution FGOC from a different perspective and propose a novel pipeline to address this issue.

B. Image Super-Resolution

It focuses on reconstructing high-resolution output from the low-resolution input. Deep learning-based SR methods have been widely studied, achieving the state of the arts, e.g., attention-based methods [35], [36], feature aggregation-based methods [37], [38], and reference-based methods [39]. Currently, some other advanced technologies have been extended to SR, such as meta-learning [40], [41], zero-shot learning [42], and generative adversarial learning [43].

In this work, we do not aim to build strong SR methods. However, we focus on exploring how to utilize SR features to achieve our feature balance strategy and iterative interaction mechanism, thereby improving the accuracy of the low-resolution FGOC.

C. Auxiliary Knowledge

Auxiliary knowledge can be adopted to improve the performance of the fine-grained classification [44]. Song *et al.* [45] leveraged auxiliary language modalities to improve the fine-grained classification. Unstructured text is used to facilitate fine-grained classification in [46]. In contrast, we utilize elaborate features from SR to improve the low-resolution FGOC.

SR can reconstruct high-resolution detailed features, providing auxiliary structure information to improve visual tasks, e.g., [14], [15], [47]. Specifically, these methods build SR-based constraint losses to train the main-task network to learn detailed features. However, the relationship transfer [15] and direct supervision [14], [47] are not sufficient for FGOC. The reason is that SR's weak features are inhibited by its strong features since they are equally dealt with. Instead, we study the feature balance strategy to guarantee that the weak detailed features from SR are efficiently integrated. Moreover, we explore the iterative interaction mechanism between SR network and FGOC network, discovering that fine-grained classification can help SR network optimize the discriminative parts of objects.

III. Method

A. Overview

SR can provide auxiliary detailed features, contributing a promising solution for the LR FGOC. However, directly integrating SR features to LR features hardly obtains the optimal solution. Since the weak features in SR may be inhibited by the strong features when they are integrated to LR features, and thus, the weak features are not utilized efficiently to improve the classification performance (see Section III-B1 for more details). Instead, we propose a feature balance strategy to efficiently leverage SR features, improving the accuracy of FGOC. Moreover, we build an iterative interaction mechanism between SR and FGOC, alternately optimizing their performances. An overview of the proposed framework is shown in Fig. 2.

1) SR Features: Given an SR dataset $\mathcal{D}_{\mathcal{S}} = \{I_{\text{lr}}, G_{\text{sr}}\}$, where I_{lr} is a low-resolution image set, having the same scale with the corresponding high-resolution image set G_{sr}, we train an SR model $\mathcal{M}_{\mathcal{S}}(W_S)$ to learn high-resolution detailed features (F_{sr}) of ships, where W_S is SR model's parameters. Notice that $\mathcal{M}_{\mathcal{S}}$ can be any existing pre-upsampling SR models. We logically divide $\mathcal{M}_{\mathcal{S}}$ into three parts: feature extraction via the first convolution layer, super-resolution feature reconstruction by a series of intermediate convolution operations, and super-resolution image generation with the last convolution layer (see Fig. 2).

2) LR Features: We train the LR model $\mathcal{M}_{\mathcal{L}}(W_L)$ to generate objects' low-resolution coarse features (F_{lr}) through restructuring the low-resolution input image, where W_L stands for LR model's parameters. $\mathcal{M}_{\mathcal{L}}$ has the same architecture with $\mathcal{M}_{\mathcal{S}}$. Especially, their feature extraction parts share the parameters to make the feature space identical.

The SR features F_{sr} and LR features F_{lr} are then leveraged to implement the proposed feature balance strategy, generating efficient combined features to feed into the classification network for FGOC. Sequentially, we design an induced super-resolution loss based on the previous ship classification results and further implement the proposed iterative interaction mechanism to optimize FGOC. The detailed descriptions are given as follows.

B. Feature Balance

SR features F_{sr} can be helpful for FGOC since they contain high-resolution detailed features, and the core issue is how to efficiently integrate the detailed features to coarse contents of the low-resolution objects. First, we need to define what are the strong and weak features. To better explain this, Fig. 3 provides a graphic. The strong/weak features in F_{sr} are defined according to their correlations with the features in F_{lr}. Specifically, if the feature in F_{sr} is highly correlated with the corresponding feature in F_{lr}, the feature in F_{sr} is defined as the strong feature [see F_1 in Fig. 3(a)]. Otherwise, if the feature in F_{sr} is lowly correlated with the corresponding feature in F_{lr}, the feature in F_{sr} is defined as the weak feature [see F_2 in Fig. 3(a)]. Mathematically, we adopt a sigmoid of pixel-wise cosine similarity to measure the correlation $\text{Cr}_{i,j}$

$$\text{Cr}_{i,j} = \text{Sig}\left(\frac{F_{\text{lr}}^{i,j} \cdot F_{\text{sr}}^{i,j}}{\left|F_{\text{lr}}^{i,j}\right|\left|F_{\text{sr}}^{i,j}\right|}\right) \tag{1}$$

where $F_{\text{sr}}^{i,j} \in \mathbb{R}^{1\times1\times C}$ and $F_{\text{lr}}^{i,j} \in \mathbb{R}^{1\times1\times C}$ stand for the (i, j)th pixel's SR feature vector and LR feature vector, respectively,

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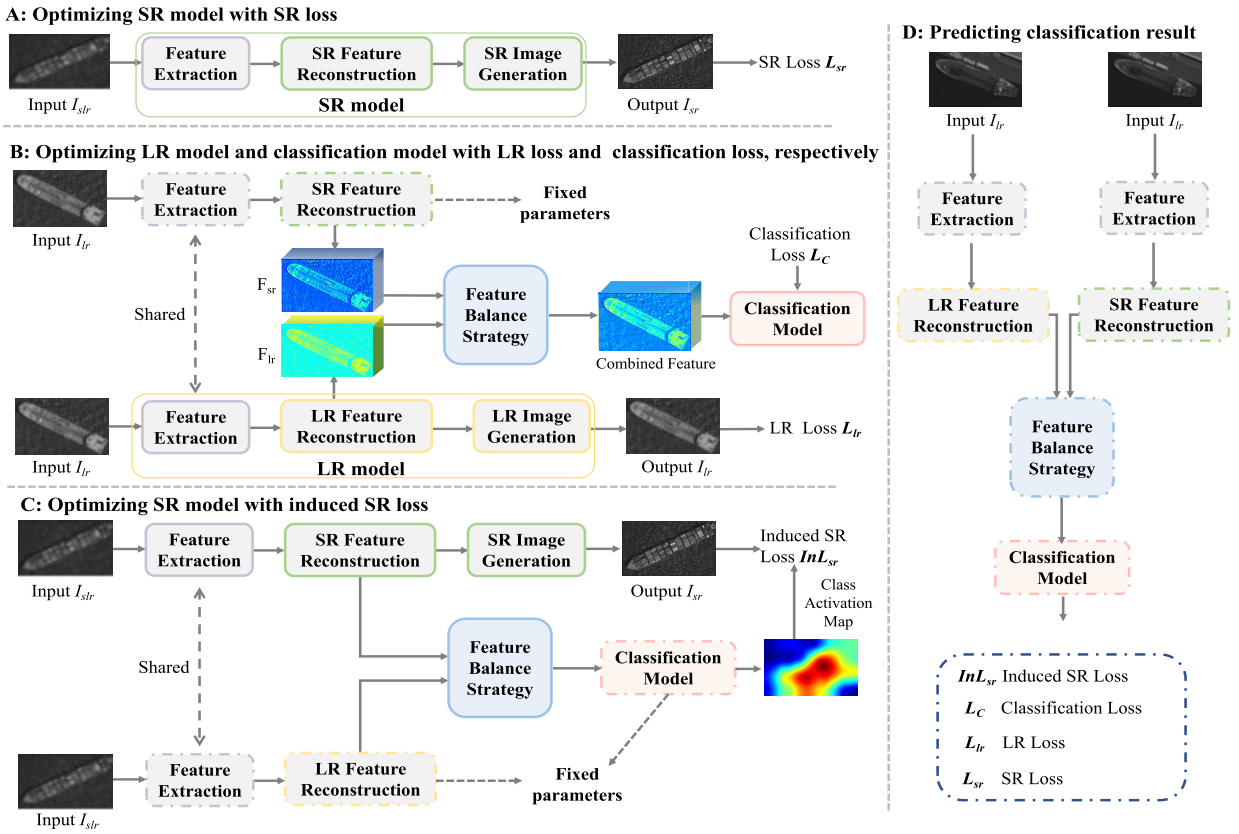


Fig. 2. Pipeline of the proposed framework. A classification model adopts ResNet50 [33] as the backbone. The SR model and LR model use RFDN [34] as the backbone, where the first convolution layer and last convolution layer serve as feature extraction and SR/LR image generation, respectively, and the other convolution layers serve as SR/LR feature reconstruction. We first optimize the SR model to generate SR features for the simulated input image I_{slr}. Then, we optimize the LR model and classification model to produce a classification result for the realistic LR image I_{lr}. After that, we structure an induced super-resolution loss to improve the SR model. The interaction is iterative to obtain the final FGOC. In the predicting classification stage, the realistic LR image I_{lr} is used to obtain the classification result.

and Sig expresses a sigmoid operation. A higher $\text{Cr}_{i,j}$ stands for a stronger feature in F_{sr}.

Sequentially, we propose a feature balance strategy by discriminatively integrating the weak and strong features of F_{sr} to the coarse features of F_{lr}, highlighting the weak features to relieving their inhibition. The structure is shown in Fig. 4(f). Specifically, if the LR features are weak or missing, the corresponding SR features (i.e., weak features) are more integrated; otherwise, if the LR features are strong, the corresponding SR features (i.e., strong features) are less integrated. This will make the combined features balance between the weak and strong features, strengthening the role of the weak features for FGOC. More precisely, as shown in (1), $F_{\text{sr}} \in \mathbb{R}^{W\times H\times C}$ is distinctively integrated to $F_{\text{lr}} \in \mathbb{R}^{W\times H\times C}$ through the inverse correlation measure-based map of $1-\text{Cr}$, where W, H, and C are the width, height, and channel of the feature, respectively, and $\text{Cr} \in \mathbb{R}^{W\times H}$ is the correlation measure map

$$F_c = F_{\text{lr}} + (1-\text{Cr}) \circledast F_{\text{sr}} \tag{2}$$

where F_c is the combined features and $\circledast$ represents the pixel-level multiplication. Fig. 5 shows the visual results for a more intuitive understanding of the feature balance process.

1) Explanation on Feature Inhibition: Existing strategies [14], [15], [47] generally treat the weak and strong features in F_{sr} equally when they are furnished to low-resolution feature F_{lr}. Thereby, the weak features will be inhibited by strong features, degrading the utilization efficiency of F_{sr}. The reason is that the weights of weak features are lower or equal to the ones of strong features, which will make that the weak feature values occupy a small proportion of the strong features after the feature integration (i.e., weak features are inhibited), as shown in Fig. 3(c). Although the weak features are integrated to F_{lr}, the strong features are equally integrated to F_{lr}. Therefore, the strong features will make the corresponding integrated features play a major role in object classification, i.e., the weak features in F_{sr} are not utilized efficiently. As shown in Fig. 3, the proportion of the weak feature F_2 increases 5% (reaches 57%) by the equal feature integration than the one in F_{lr} [see Fig. 3(b) and (c)]. In contrast, our method obtains a better feature balance, which increases 41% (reaches 93%) than the one in F_{lr} [see Fig. 3(d)].

Moreover, we design five feature integration alternatives, i.e., sum, concat, attention weighting, SR transfer [14], and relationship transfer [15], as shown in Fig. 4(a)–(e). Their mathematical descriptions are given as follows. Besides, the detailed evaluations of these alternatives are provided in Section V-C.

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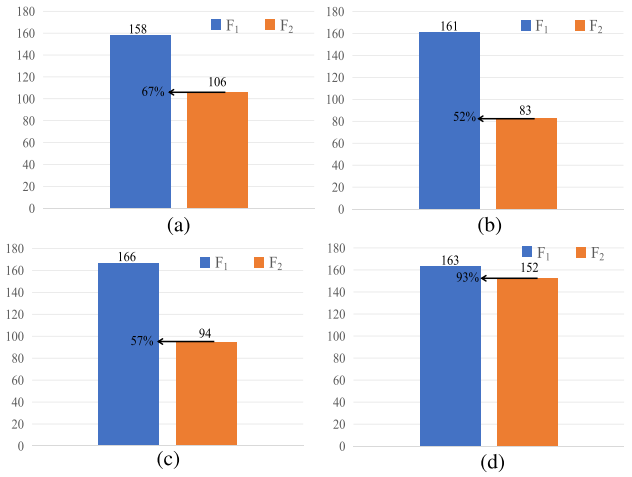


Fig. 3. Graphic to explain the strong and weak features, and feature inhibition. The feature F_1 in F_{sr} is highly correlated with the feature F_1 in F_{lr}, which is defined as the strong feature. On the other hand, the feature F_2 in F_{sr} is lowly correlated with the feature F_2 in F_{lr}, which is defined as the weak feature. (a) Feature values in Fsr. (b) Feature values in Flr. (c) Equal feature integration. (d) Discriminative feature integration.

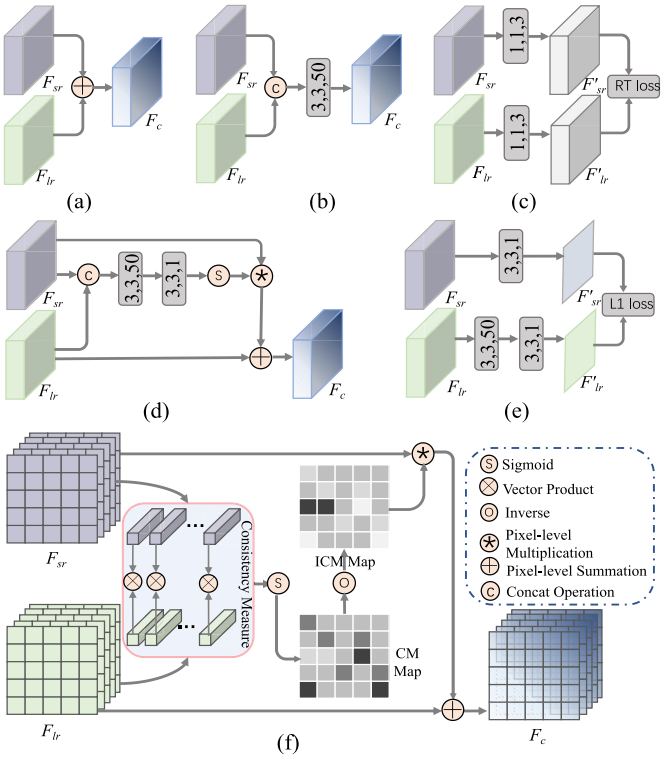


Fig. 4. Architectures of different feature integration alternatives. "h, w, and c" in the rectangular box stand for the height, width, and channel of the convolution kernel, respectively. (a) Feature combination via adding. (b) Feature combination via concatenating. (c) Supervising LR features via relationship transfer (RT) loss [15]. (d) SR features integrated via attention weighting. (e) Supervising LR features [14] via L_1 loss. (f) Feature balance strategy. l, w, and c stand for the length, width, and channel number of the convolution kernel, respectively.

a) Feature adding: The combined feature F_c is obtained by adding F_{lr} and F_{sr}, which can be formulated as

$$F_c = F_{\text{lr}} \oplus F_{\text{sr}} \tag{3}$$

where $\oplus$ stands for pixel-wise summation.

b) Feature concatenating: The combined feature F_c is obtained by concatenating F_{lr} and F_{sr}, which is formulated as

$$F_c = \text{CB}(F_{\text{lr}} \uplus F_{\text{sr}}) \tag{4}$$

where $\uplus$ denotes the concatenating operation and CB stands for a convolution block.

c) Relationship transfer (RT): LR feature is improved by transferring the context relationship of SR feature, where RT loss [15] is adopted

$$L_{\text{RT}} = \frac{1}{W^2H^2}\sum_{i=1}^{\text{WH}}\sum_{j=1}^{\text{WH}}||S_{\text{sr}}(i,j) - S_{\text{lr}}(i,j)||_1 \tag{5}$$

$$S(i,j) = \left(\frac{F'(i)}{||F'(i)||_2}\right)^T\left(\left(\frac{F'(j)}{||F'(j)||_2}\right)\right) \tag{6}$$

where S_{sr} and S_{lr} are the SR feature similarity matrix and LR feature similarity matrix, respectively, F' stands for a one-channel feature, and i/j is the pixel position index.

d) Attention weighting: The combined feature F_c is achieved by attention weighting of F_{lr} and F_{sr}, which is defined as

$$F_c = F_{\text{lr}} \oplus (F_{\text{sr}} \circledast Sig(CB(F_{\text{sr}} \uplus F_{\text{lr}}))). \tag{7}$$

e) SR transfer: LR feature is improved by transferring the knowledge of SR feature, which uses the L_1 loss

$$L_1 = ||\text{CB}(F_{\text{sr}}) - \text{CB}(F_{\text{lr}})||_1 \tag{8}$$

where $||\cdot||_1$ expresses the 1-norm operation.

C. Iterative Interaction

A fine-grained object classification model tends to utilize the most discriminative regions of objects. Therefore, we can induce the SR network to pay more attention to these regions, pertinently optimizing corresponding high-resolution features. Specifically, we introduce the iterative interaction to improve SR and classification performances. First, we utilize previous object classification network to generate the class activation maps. Then, we structure an induced super-resolution loss to refine the SR network. After that, the refined SR network helps classification model improve performance. The interaction is iterative T times to obtain the final FGOC. The proposed induced loss $\text{InL}_{\text{sr}}(I_{\text{slr}}; G_{\text{sr}})$ can be written as

$$\text{InL}_{\text{sr}}(I_{\text{slr}}; G_{\text{sr}}) = \min_{W_S^t}\frac{1}{N}\sum_{n=1}^{N}\text{GC}_n\left(\mathcal{M}_C\left(W_C^{t-1}\right)\right) \circledast \left|I_{\text{slr}}^n - G_{\text{sr}}^n\right| \tag{9}$$

where I_{slr}^n and G_{sr}^n are the nth input simulated LR image and SR ground truth, respectively, N is the total number of the image pairs, and GC_n stands for the nth image's class activation map obtained by the Grad-CAM operation [16]. W_S^t is weight parameters of the SR model in the $t \in T$th interaction. $\mathcal{M}_C(W_C^{t-1})$ expresses the object classification model $\mathcal{M}_C$ with the t-1th interaction parameters of W_C.

Then, these high-resolution features are provided to LR features, achieving a better feature balance to improve the performance of FGOC. We alternate the fine-grained ship

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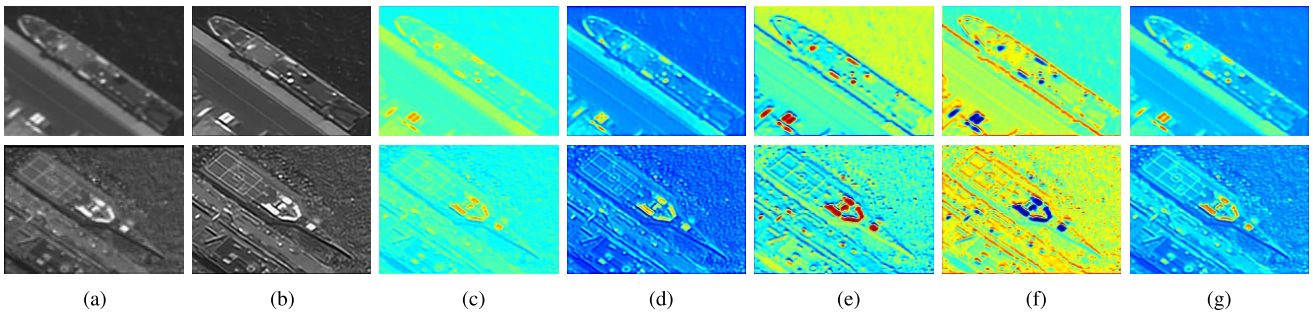


Fig. 5. Visual results in the process of the feature balance. (a) LR image. (b) SR image. (c) LR feature. (d) SR feature. (e) Feature CM map. (f) Inverse feature consistency measure (ICM) map. (g) Combined feature. Only if LR features are weak, ICM will generate large weighting values and the SR features are more integrated; otherwise, if LR features are strong, SR features will be less integrated. Thus, the weak and strong presentations in the combined feature achieve a relative balance, avoiding that the weak presentations are inhibited by the strong ones.

Algorithm 1 Optimization Process of our Framework

Input: Training datasets $\mathcal{D}_{\mathcal{C}} = \{I_{\text{lr}}, G_c\}$ from LFS and $\mathcal{D}_{\mathcal{S}} = \{I_{slr}, G_{\text{sr}}\}$ from SSR.
Stage 1: Training SR model, LR model and FGOC model.
1. Optimizing SR model using Equation (10).
2. Jointly optimizing LR model and FGOC model using Equation (11).
Stage 2: Iterative interactive optimization.
3. **While** optimization is not convergent **do**
4. Optimizing induced SR model using Equation (9).
5. Jointly optimizing LR model and FGOC model using Equation (11).
6. **End while**
Output: W_S, W_L and W_C.

classification and induced SR several times to gradually refine FGOC. This is different from the strategy of iterating SR and classification models at the same time. Our iterative interaction permits SR and classification models to be optimized using their corresponding datasets of SSR and LFS, respectively. However, simultaneously iterating SR and classification models can only use the dataset of SSR, which may limit the classification model' performance on realistic low-resolution aerial images.

D. Optimization

Given training datasets $\mathcal{D}_{\mathcal{C}} = \{I_{\text{lr}}, G_c\}$ from LFS and $\mathcal{D}_{\mathcal{S}} = \{I_{\text{slr}}, G_{\text{sr}}\}$ from SSR, where I_{lr} and G_c are the LR image set and corresponding classification ground truth, respectively, and I_{slr} and G_{sr} are the simulated LR image set and corresponding SR ground truth, respectively, the optimization process of our framework includes two stages, as illustrated in Algorithm 1. In stage 1, the SR model is trained to learn weight parameters W_S with loss L_{sr}

$$L_{\text{sr}}(I_{\text{slr}}; G_{\text{sr}}) = \min_{W_S} \frac{1}{N} \sum_{n=1}^{N} \left| I_{\text{slr}}^n - G_{\text{sr}}^n \right|. \tag{10}$$

Then, the LR model and FGOC model are jointly optimized to learn weight parameters W_C and W_L with cross-entropy

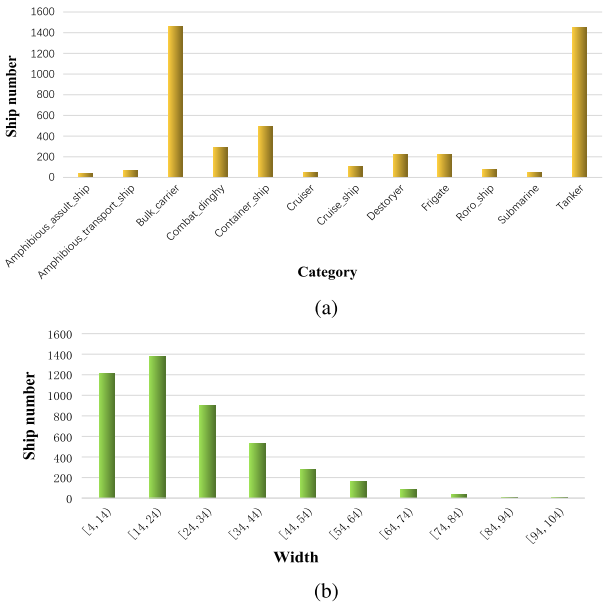


Fig. 6. Statistics of the proposed low-resolution fine-grained ship dataset (LFS). (a) Ship subcategory statistic. (b) Ship scale statistic.

classification loss L_C and LR reconstruction loss L_{lr}

$$L_{\text{joint}}(I_{\text{lr}}; G_c, G_{\text{lr}}) = \min_{W_C, W_L} (L_C(I_{\text{lr}}; G_c) + L_{\text{lr}}(I_{\text{lr}}; G_{\text{lr}})) \tag{11}$$

$$L_C(I_{\text{lr}}; G_c) = \min_{W_C} -\frac{1}{N'} \sum_{n=1}^{N'} G_c^n \log\left(P\left(I_{\text{lr}}^n\right)\right) \tag{12}$$

$$L_{\text{lr}}(I_{\text{lr}}; G_{\text{lr}}) = \min_{W_L} \frac{1}{N'} \sum_{n=1}^{N'} \left| I_{\text{lr}}^n - R\left(I_{\text{lr}}^n\right) \right| \tag{13}$$

where N' is the total number of LR images, $P(I_{\text{lr}}^n)$ expresses the output probability of the FGOC model for the nth input image, and $R(I_{\text{lr}}^n)$ is the output reconstruction image of the LR model for the nth input image I_{lr}. In stage 2, we first obtain GC with the previous ship classification and then optimize the SR model by the induced loss InL_{sr}. After that, we jointly optimize the LR model and FGOC model using L_{lr} and L_C. The interactive optimization is alternated several times to generate the final FGOC result.

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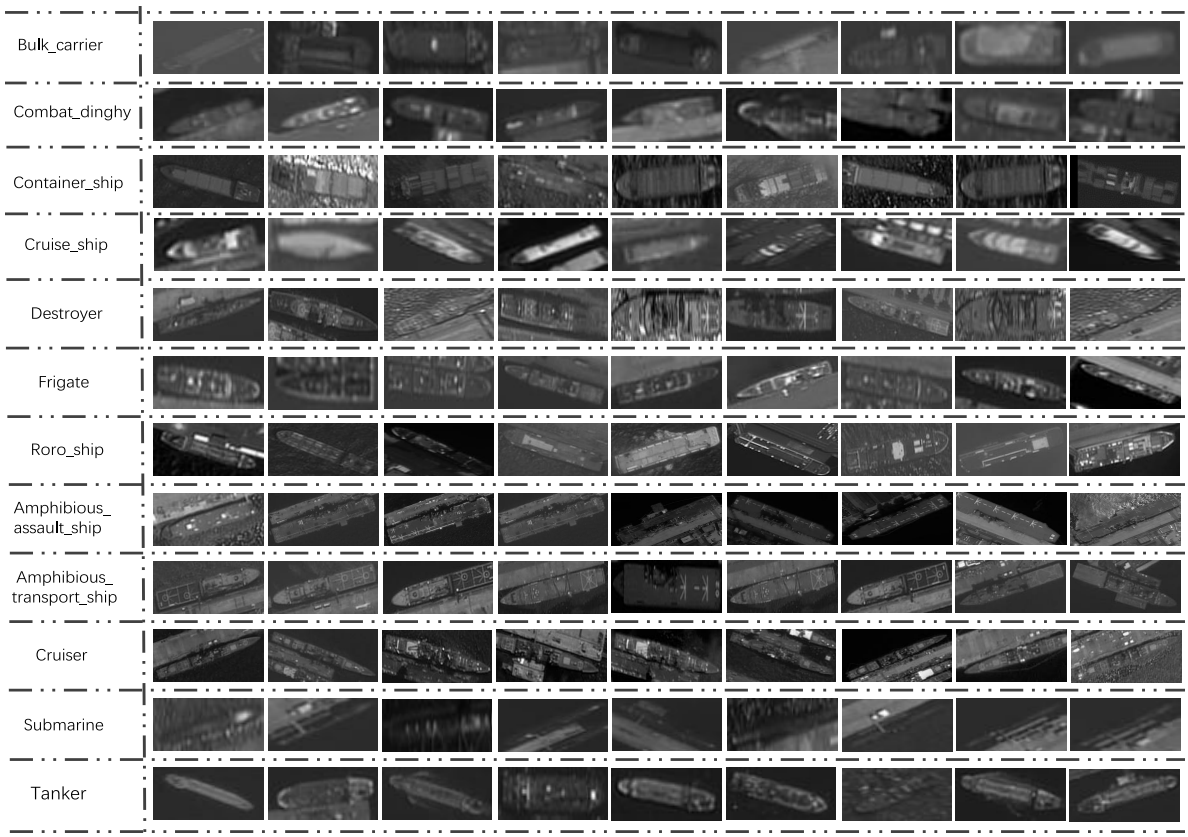


Fig. 7. Visual samples of each subcategory in the proposed low-resolution fine-grained object dataset (LFS).

IV. Proposed Datasets

In order to support the study of the low-resolution FGOC in optical aerial images, we build two datasets: one is the low-resolution fine-grained ship dataset (LFS) and the other one is a ship super-resolution dataset, named SSR. Their desirability and details are elaborated as follows.

A. LFS

Current optical aerial datasets for FGOC primarily reveal two deficiencies: The first deficiency is the limited ship subcategories, e.g., two types of ships and nonships in MASATI [17] and four ship subcategories in BCCT200 [18]; The second deficiency is the ship scales' unbalance that are mainly concentrated in large ones. In contrast, our LFS mainly includes small-scale ships with less details and covers 12 subcategories with military ships and civil ships. Specifically, the ship images collected from GF-1 and GF-2 are panchromatic, and the total number is 4591 where 3217 images for training and 1374 images for testing respectively. Each ship subcategory is labeled by professionals. The detailed statistics are shown in Fig. 6. Especially, the number of ships with width range of 4–24 pixels is 56% in LFS. Fig. 7 shows some representative visual samples.

B. SSR

Existing super-resolution datasets commonly focus on the visible-light natural scenes, lacking the pertinence of ship super-resolution on the optical aerial images. To effectively train the ships' SR model, we build a ship super-resolution dataset (SSR). Specifically, we first collect 979 high-resolution ship images from GF-1 and GF-2. After that, each high-resolution ship image is downsampled by 2, 3, 4, 5, and 6 times with the bicubic interpolation operation,[2] and then are upsampled to the original scale as the corresponding low-resolution ship image. Therefore, the SSR dataset contains 4895 low- and high-resolution pairs, where 3916 pairs are for training and 979 pairs are for testing, respectively.

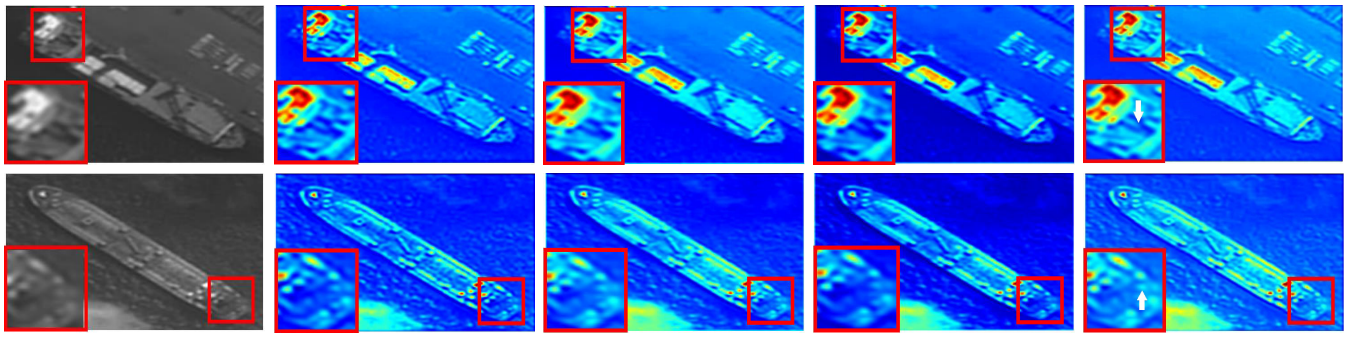
V. Experiments

A. Datasets

We conduct extensive experiments on the proposed LFS dataset and on the other three datasets of FS23 [2], HRSC2016 [19], and DOTA [48]. LFS includes 4591 ship images split into 3217 training and 1374 testing images, which covers 12 subcategories focused on low resolution. FS23 mainly contains high-resolution ship images with the total number of 4081 split into 3256 training and 825 testing images, which is divided into 23 subcategories. HRSC2016 comprises 2149 ship images split into 1220 training and 929 testing images with 24 subcategories, and most of them are high resolution. DOTA includes 15 subcategories, and the samples with resolution $< 50 \times 50$ pixels are selected, i.e., 58 284 training and 16 370 testing samples are used in our

[2]Bicubic interpolation is implemented using library function $I_o = \text{imresize}(I_i, [W/f, H/f], '\text{bicubic}')$ in MATLAB R2018b, where I_i is the input image with the size of $W \times H$, f is a reduction factor, and I_o is the downsampled image.

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(a) (b) (c) (d) (e)

Fig. 8. Visual comparisons of different feature combination strategies. (a) Input low-resolution image. (b) Attention weighting strategy. (c) Supervising low-resolution features. (d) Relationship transfer strategy. (e) Feature balance strategy. Our feature balance strategy can help combined features retain more weak detail features (see the regions pointed by white arrows).

experiment. The parameters of the models in the experiments are optimized on the training datasets and are selected and validated on the corresponding testing datasets.

B. Implementation Details

Our framework includes three submodels: 1) SR model generates SR features, which is fully supervised trained with dataset SSR; 2) LR model extracts the features of realistic low-resolution aerial images, which is self-supervised trained by restructuring the input image on dataset LFS (the advantage is that it can help extract the features of low-resolution images without using additional labels); and 3) classification model distinguishes object based on its SR and LR feature integration, which is fully supervised trained with dataset LFS. We adopt ResNet50 [33] pretrained on ImageNet as the classification backbone for a fair comparison with existing works (e.g., [2], [21], [49], [50]). For both SR model and LR model, we use a lightweight RFDN [34] as the backbone. All experiments are conducted on a GTX 1080Ti GPU using PyTorch. SGD with the momentum of 0.9 and weight decay of $1e^{-4}$ and Adam with the momentum of 0.9 are used to optimize the classification network and SR network, respectively. The batch size is set to 8. The input images are resized to 128×256 pixels. Here, the parameters of pretrained convolution blocks in ResNet50 are used as the initialization parameters to overcome the inconsistent input sizes.

In stage 1, an initial learning rate of 0.001 is adopted in the first 30 epochs for the SR model, LR model, and classification model and then is divided by 10 for the next 200 epochs. Following this, in stage 2, we train these models using a learning rate of 0.0001, and each interaction's training epoch is 200.

C. Ablation Analysis

1) Effect of Feature Balance Strategy: In Section III-B, we propose the feature balance strategy by discriminatively integrating the weak and strong features of F_{sr} to coarse features of F_{lr}, avoiding that the weak features are inhibited by strong features. Moreover, we build five alternatives for comparisons: 1) feature combination via adding; 2) feature combination via concatenating; 3) SR features integrated via attention weighting; 4) supervising LR features via SR image [14]; and 5) supervising LR features via relationship transfer of SR features [15]. Their detailed structures are shown in Fig. 4. Table I reports the comparison results. SR features can improve the accuracy of FGOC (e.g., the alternative 1) increases the accuracy by 2.4% than the baseline). However, the alternatives of 1)–5) equally treat the weak and strong features of SR when they are combined to LR features where the weak features are restrained by strong features, limiting their performances. Our proposed feature balance strategy achieves the best performance with the accuracy of 89.5% since it can efficiently help balance the weak and strong features when structuring the combined features.

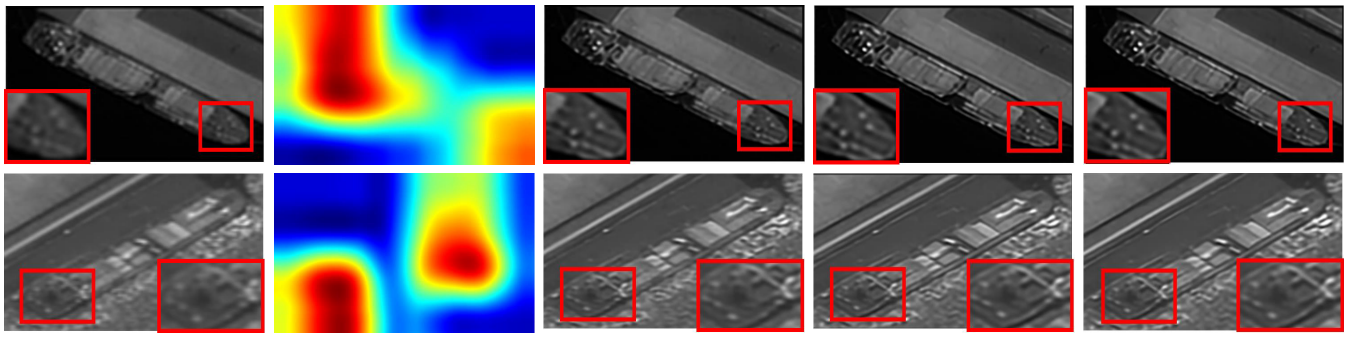
TABLE I
EFFECT STUDY OF THE FEATURE BALANCE STRATEGY WITH CLASSIFICATION MEASURE *Accuracy* ON THE LFS DATASET

Different alternatives	**Accuracy (%)**
(a) Feature combination via adding	87.1
(b) Feature combination via concatenating	86.5
(c) SR features integrated via attention weighting	87.9
(d) Supervising LR features via SR image	87.3
(e) Supervising LR features via relationship transfer	88.2
(f) Baseline with ResNet50	84.7
(g) Feature balance strategy (Ours)	**89.5**

TABLE II
EFFECT STUDY OF THE ITERATIVE INTERACTION MECHANISM WITH SUPER-RESOLUTION MEASURE PSNR AND CLASSIFICATION MEASURE *Accuracy* ON LFS THE DATASET

Iterative interaction times T	**PSNR**	**Accuracy (%)**
Interaction time $T = 0$	30.3	85.2
Interaction time $T = 1$	31.0	89.5
Interaction time $T = 2$	32.4	90.2
Interaction time $T = 3$	**32.5**	**90.5**

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(a) (b) (c) (d) (e)

Fig. 9. SR results in the process of the iterative interaction. (a) Input LR image. (b) Class activation map. (c) SR result of the first interaction. (d) SR result of the second interaction. (e) SR result of the third interaction. SR results are gradually improved as the number of interactions increases.

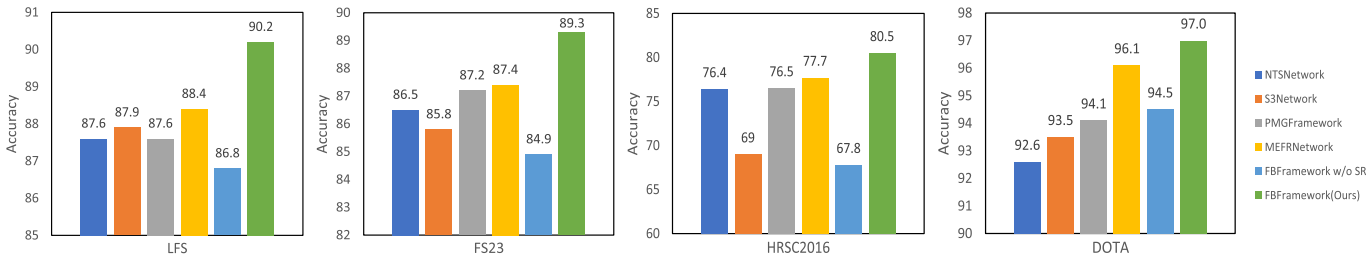


Fig. 10. Comparison of our method with recent state-of-the-art methods on LFS, FS23, HRSC2016, and DOTA datasets using Top-1 Accuracy (%).

TABLE III

TOP-1 ACCURACY (%) RESULTS OF DIFFERENT METHODS ON THE TESTING SET OF LFS DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED IN THE TABLE, AND THE BEST ACCURACY RATE IS BOLDED

Category ID	PMGFramework	MEFRNetwork	FBFramework without SR	FBFramework (Ours)
01. Amphibious assault ship	**100.0**	**100.0**	**100.0**	**100.0**
02. Amphibious transport ship	**88.0**	71.4	76.2	76.2
03. Bulk carrier	53.3	**90.4**	84.7	89.3
04. Combat dinghy	97.3	**98.9**	85.4	91.0
05. Container ship	85.7	89.3	**90.0**	86.7
06. Cruiser	86.1	40.0	**86.7**	**86.7**
07. Cruise ship	64.0	**79.4**	**79.4**	76.5
08. Destoryer	**89.3**	69.6	81.2	87.0
09. Frigate	**100.0**	71.6	76.1	80.6
10. Roro ship	73.5	84.0	80.0	**88.0**
11. Submarine	88.4	40.0	86.7	**100.0**
12. Tanker	76.1	94.3	92.2	**95.9**
Overall Accuracy	87.6	88.4	86.8	**90.2**

2) Visualization: We visualize the combined features of three representative strategies (i.e., attention weighting, supervising LR features [14], and relationship transfer [15]) and our feature balance strategy, as shown in Fig. 8. The detail features are weak since they are inhibited by strong features [see the enlarged red rectangle regions in Fig. 8(b)–(d)]. In contrast, our feature balance strategy persists more detail features through restraining the strong features while highlighting the weak features, as shown in the regions pointed by white arrows in Fig. 8(e).

3) Effect of Iterative Interaction Mechanism: In Section III-C, we implement an iterative interaction mechanism to improve the performance of FGOC. On the one hand, we utilize the class activation maps generated by the classification results to induce the SR network to refine the most discriminative regions of ships. On the other hand, these refined SR features further help the classification network improve the accuracy of FGOC. Here, we study the influence of different iterative interaction times. As shown in Table II, the performances of both SR and classification are improved

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TABLE IV

TOP-1 ACCURACY (%) RESULTS OF DIFFERENT METHODS ON THE TESTING SET OF DOTA DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED IN THE TABLE, AND THE BEST ACCURACY RATE IS BOLDED

Category ID	PMGFramework	MEFRNetwork	FBFramework without SR	FBFramework (Ours)
01. Baseball-diamond	8.8	**97.1**	0.0	76.4
02. Basketball-court	78.2	**98.4**	86.9	86.9
03. Bridge	**99.3**	95.8	91.4	96.7
04. Ground-track-field	42.8	**94.4**	14.2	85.7
05. Harbor	90.3	**97.9**	91.8	97.6
06. Helicopter	82.4	**95.8**	27.2	63.6
07. Large-vehicle	73.5	**97.8**	82.5	88.0
08. Plane	84.2	**99.1**	91.7	98.6
09. Roundabout	80.9	**93.2**	71.4	81.2
10. Ship	90.2	96.4	**98.8**	98.5
11. Small-vehicle	18.1	93.2	97.9	**98.2**
12. Soccer-ball-field	78.5	80.3	54.7	**80.9**
13. Storage-tank	95.0	95.6	90.3	**97.5**
14. Swimming-pool	69.6	**91.7**	85.9	90.8
15. Tennis-court	**97.5**	96.1	76.4	64.7
Overall Accuracy	94.1	96.1	94.5	**97.0**

TABLE V

TOP-1 ACCURACY (%) RESULTS OF DIFFERENT METHODS ON THE TESTING SET OF FS23 DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED IN THE TABLE, AND THE BEST ACCURACY RATE IS BOLDED

Category ID	PMGFramework	MEFRNetwork	FBFramework without SR	FBFramework (Ours)
01. Non ship	87.6	90.7	87.6	**92.8**
02. Aircraft carrier	83.3	**97.1**	88.2	94.1
03. Destoryer	**91.7**	84.3	87.0	**91.7**
04. Landing craft	**100.0**	81.8	90.9	95.5
05. Frigate	96.6	**98.3**	93.2	94.9
06. Amphibious transport dock	68.9	**94.4**	55.6	77.8
07. Cruiser	84.2	**91.5**	84.7	89.8
08. Tarawa-class amphibious assult ship	**100.0**	88.9	88.9	94.4
09. Amphibious assult ship	**100.0**	**100.0**	83.9	**100.0**
10. Command ship	72.5	66.7	88.9	**94.4**
11. Submarine	84.8	**95.8**	**95.8**	93.8
12. Medical ship	91.2	**100.0**	**100.0**	**100.0**
13. Combat boat	60.0	93.1	93.1	**100.0**
14. Auxiliary ship	**83.3**	66.7	62.2	68.9
15. Container ship	**95.0**	85.0	80.0	85.0
16. Car carrier	90.9	92.9	92.9	**100.0**
17. Hover craft	94.4	**100.0**	**100.0**	**100.0**
18. Bulk carrier	**100.0**	78.3	71.0	78.3
19. Oil tanker	**94.9**	87.9	84.8	84.8
20. Fishing boat	**77.8**	30.0	70.0	60.0
21. Passenger ship	86.4	**88.9**	72.2	66.7
22. Liquefied gas ship	88.9	**100.0**	**100.0**	**100.0**
23. Barge	93.5	**100.0**	**100.0**	**100.0**
Overall Accuracy	87.2	87.4	84.9	**89.3**

with the increase of interaction time T. When $T \geqslant 2$, the performances tend to be stable. Comprehensively considering both optimization time and performance, we take T to 2 in our experiments.

4) Visualization: Fig. 9 shows the visual results. Class activation maps in Fig. 9(b) reveal the most discriminative regions of ships. Then, our iterative interaction mechanism gradually refines the SR results, especially on these

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TABLE VI

TOP-1 ACCURACY (%) RESULTS OF DIFFERENT METHODS ON THE TESTING SET OF HRSC2016 DATASET. THE AVERAGE ACCURACY AND ACCURACY OF EACH CLASS ARE LISTED IN THE TABLE, AND THE BEST ACCURACY RATE IS BOLDED

Category ID	PMGFramework	MEFRNetwork	FBFramework without SR	FBFramework (Ours)
01. Ship	71.3	68.9	51.5	**78.4**
02. Warcraft	**15.4**	7.7	7.7	7.7
03. Nimitz class aircraft carrier	**84.4**	69.6	69.6	78.3
04. Enterprise class aircraft carrier	81.6	50.0	60.0	**90.0**
05. Arleigh Burke class destroyers	46.7	79.0	76.6	**86.3**
06. WhidbeyIsland class landing craft	66.7	68.3	63.4	**75.6**
07. Perry class frigate	33.3	94.6	75.7	**95.5**
08. Sanantonio class amphibious transport dock	**81.8**	41.2	47.1	70.6
09. Ticonderago class cruiser	81.8	86.2	78.7	**88.3**
10. Admiiral Kuznetsov aircraft carrier	**76.1**	0.0	50.0	50.0
11. Austen class amphibious transport dock	50.0	75.0	**78.1**	**78.1**
12. Tarawa-class amphibious assault ship	80.6	**97.4**	73.7	86.8
13. Container ship	**78.3**	60.0	46.7	53.3
14. Oxo	50.0	63.6	63.6	**69.7**
15. Car carrier	**100.0**	0.0	33.3	33.3
16. Hovercraft	76.9	**100.0**	84.1	88.6
17. Yacht	87.5	**100.0**	84.1	88.6
18. Cargo ship	70.0	83.0	**84.1**	77.3
19. Cruise ship	**87.1**	25.0	25.0	25.0
20. Sumarine	70.7	**88.9**	69.4	86.1
21. Lute	**91.9**	0.0	50.0	50.0
22. Medical ship	58.8	**100.0**	70.0	**100.0**
23. Transport ship	**81.9**	61.5	76.9	76.9
24. Midway-class aircraft carrier	50.0	75.0	87.5	**100.0**
Overall Accuracy	76.5	77.7	67.8	**80.5**

TABLE VII

TOP-1 ACCURACY (%) RESULTS OF CROSS ASSESSMENTS ON THE TESTING SET OF FS23 AND HRSC2016 DATASETS. $X \rightarrow Y$ STANDS FOR THE MODEL THAT IS TRAINED ON X AND TESTED ON Y

Category	FS23→ FS23	FS23→ HRSC2016	HRSC2016→HRSC2016	HRSC2016→FS23
Landing craft	95.5	100.0	75.6	90.9
Frigate	94.9	99.1	95.5	88.1
Cruiser	89.8	96.8	88.3	81.4
Medical ship	100.0	100.0	100.0	100.0
Hover craft	100.0	100.0	88.6	79.2

discriminative regions, as shown in the enlarged red rectangle regions in Fig. 9(c)–(e). At the same time, the accuracy of FGOC is improved with the help of better SR results.

D. Comparison With the State of the Art

We compare the proposed feature balance framework (FBFramework) with four state-of-the-art methods, including multilevel enhanced feature representation network (MEFRNetwork) [2], progressive multigranularity training framework (PMGFramework) [49], selective sparse sampling network (S3Network) [21] and navigator–teacher–scrutinizer network (NTSNetwork) [50]. We train the available deep models provided by authors and recommended parameter settings for evaluating on the LFS dataset, FS23 dataset, and HRSC2016 dataset.

Fig. 10 shows the result comparisons on LFS, FS23, HRSC2016, and DOTA datasets. Our method continuously produces the best performances on these four datasets. Taking the results on LFS dataset as an example, we analyze the performance comparison in detail. The baseline model (FBFramework without SR) achieves 86.8%. Our method further improves the baseline model by 3.4%, thereby achieving the best performance against other methods. In particular, compared with MEFRNetwork [2] which enhances multilevel visual feature representations and then weights regional features to focus more on the silent regions, we achieve a better

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result with the accuracy improvement of 1.8%. Besides, our method outperforms PMGFramework [49] where the most discriminative granularities and cross multigranularity fusion are explored by 2.6%.

Moreover, detailed classification comparisons of each subcategory on four datasets are provided in Tables III, IV, V, and VI. Our method achieves considerable improvements compared with other methods.

E. Cross Assessment

Our framework also works for new unseen images. To verify this, we train our model on FS23 [2] and test on HRSC2016 [19]. On the other hand, we train our model on HRSC2016 and test on FS23. As shown in Table VII, our framework not only performs well on the own test dataset but also achieves good results on the other test dataset.

VI. Conclusion

We explore the low-resolution fine-grained object classification. In particular, we propose a feature balance strategy to discriminatively integrate high-resolution weak and strong features to low-resolution coarse features, highlighting the weak features to relieving their inhibition. In addition, we further implement an iterative interaction mechanism to alternately refine feature details of the discriminative object regions and optimize the performance of FGOC. Extensive experimental results illustrate that the proposed method achieves better performance than other state-of-the-art approaches. Moreover, we construct a low-resolution fine-grained object dataset, and we believe that this dataset will promote further research and evaluation of low-resolution fine-grained object classification. Especially, we implement a feature balance strategy to integrate SR and LR features to improve classification accuracy. Thus, the SR features are needed to be extracted in the inference stage, which is time-consuming. Further study focuses on the knowledge distillation method that directly teaches the LR branch to restructure SR features to relieve this issue.

References

[1] W. Li *et al.*, "Integrated localization and recognition for inshore ships in large scene remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 936–940, Jun. 2017.

[2] X. Zhang, Y. Lv, L. Yao, W. Xiong, and C. Fu, "A new benchmark and an attribute-guided multilevel feature representation network for fine-grained ship classification in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1271–1285, 2020.

[3] G. Sumbul, R. G. Cinbis, and S. Aksoy, "Multisource region attention network for fine-grained object recognition in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4929–4937, Jul. 2019.

[4] C. M. Ward, J. Harguess, and C. Hilton, "Ship classification from overhead imagery using synthetic data and domain adaptation," in *Proc. OCEANS MTS/IEEE Charleston*, Oct. 2018, pp. 1–5.

[5] R. Ji *et al.*, "Attention convolutional binary neural tree for fine-grained visual categorization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10468–10477.

[6] H. Zheng, J. Fu, Z.-J. Zha, J. Luo, and T. Mei, "Learning rich part hierarchies with progressive attention networks for fine-grained image recognition," *IEEE Trans. Image Process.*, vol. 29, pp. 476–488, 2019.

[7] X.-S. Wei, C.-W. Xie, J. Wu, and C. Shen, "Mask-CNN: Localizing parts and selecting descriptors for fine-grained bird species categorization," *Pattern Recognit.*, vol. 76, pp. 704–714, Apr. 2018.

[8] Z. Huang and Y. Li, "Interpretable and accurate fine-grained recognition via region grouping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8662–8672.

[9] W. Ge, X. Lin, and Y. Yu, "Weakly supervised complementary parts models for fine-grained image classification from the bottom up," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3034–3043.

[10] A. Dubey, O. Gupta, R. Raskar, and N. Naik, "Maximum-entropy fine grained classification," in *Proc. 32nd Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 635–645.

[11] G. Sun, H. Cholakkal, S. Khan, F. Khan, and L. Shao, "Fine-grained recognition: Accounting for subtle differences between similar classes," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 12047–12054.

[12] D. Chang *et al.*, "The devil is in the channels: Mutual-channel loss for fine-grained image classification," *IEEE Trans. Image Process.*, vol. 29, pp. 4683–4695, 2020.

[13] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[14] R. Yan and Y. Huang, "PlugNet: Degradation aware scene text recognition supervised by a pluggable super-resolution unit," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 1–17.

[15] L. Wang, D. Li, Y. Zhu, L. Tian, and Y. Shan, "Dual super-resolution learning for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3774–3783.

[16] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[17] A.-J. Gallego, A. Pertusa, and P. Gil, "Automatic ship classification from optical aerial images with convolutional neural networks," *Remote Sens.*, vol. 10, no. 4, p. 511, Mar. 2018.

[18] K. Rainey and J. Stastny, "Object recognition in ocean imagery using feature selection and compressive sensing," in *Proc. IEEE Appl. Imag. Pattern Recognit. Workshop (AIPR)*, Oct. 2011, pp. 1–6.

[19] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. Int. Conf. Pattern Recognit. Appl. Methods*, vol. 2, 2017, pp. 324–331.

[20] H. Zheng, J. Fu, Z.-J. Zha, and J. Luo, "Looking for the devil in the details: Learning trilinear attention sampling network for fine-grained image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5012–5021.

[21] Y. Ding, Y. Zhou, Y. Zhu, Q. Ye, and J. Jiao, "Selective sparse sampling for fine-grained image recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6599–6608.

[22] L. Zhang, S. Huang, W. Liu, and D. Tao, "Learning a mixture of granularity-specific experts for fine-grained categorization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8331–8340.

[23] C. Liu, H. Xie, Z.-J. Zha, L. Ma, L. Yu, and Y. Zhang, "Filtration and distillation: Enhancing region attention for fine-grained visual categorization," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 11555–11562.

[24] Z. Wang, S. Wang, H. Li, Z. Dou, and J. Li, "Graph-propagation based correlation learning for weakly supervised fine-grained image classification," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 12289–12296.

[25] S. Min, H. Yao, H. Xie, Z.-J. Zha, and Y. Zhang, "Multi-objective matrix normalization for fine-grained visual recognition," *IEEE Trans. Image Process.*, vol. 29, pp. 4996–5009, 2020.

[26] Z. Huang, W. Li, X.-G. Xia, X. Wu, Z. Cai, and R. Tao, "A novel nonlocal-aware pyramid and multiscale multitask refinement detector for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

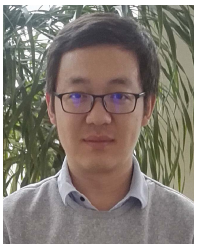
[27] Y. Han, X. Yang, T. Pu, and Z. Peng, "Fine-grained recognition for oriented ship against complex scenes in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[28] Z. Huang, W. Li, X.-G. Xia, H. Wang, F. Jie, and R. Tao, "LO-DET: Lightweight oriented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[29] L. Li, C. Wang, H. Zhang, and B. Zhang, "SAR image ship object generation and classification with improved residual conditional generative adversarial network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:43:52 UTC from IEEE Xplore. Restrictions apply.

[30] G. Sumbul, R. G. Cinbis, and S. Aksoy, "Fine-grained object recognition and zero-shot learning in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 770–779, Feb. 2018.
[31] X. Li, B. Jiang, S. Wang, L. Shen, and Y. Fu, "A human–computer fusion framework for aircraft recognition in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 297–301, Feb. 2020.
[32] J. Ji, T. Zhang, L. Jiang, W. Zhong, and H. Xiong, "Combining multilevel features for remote sensing image scene classification with attention model," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 9, pp. 1647–1651, Sep. 2020.
[33] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[34] J. Liu, J. Tang, and G. Wu, "Residual feature distillation network for lightweight image super-resolution," 2020, *arXiv:2009.11551*.
[35] X. Song *et al.*, "Channel attention based iterative residual learning for depth map super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5631–5640.
[36] B. Niu *et al.*, "Single image super-resolution via a holistic attention network," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 191–207.
[37] J. Liu, W. Zhang, Y. Tang, J. Tang, and G. Wu, "Residual feature aggregation network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2359–2368.
[38] Y. Qiu, R. Wang, D. Tao, and J. Cheng, "Embedded block residual network: A recursive restoration model for single-image super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4180–4189.
[39] G. Shim, J. Park, and I. S. Kweon, "Robust reference-based super-resolution with similarity-aware deformable convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8425–8434.
[40] S. Park, J. Yoo, D. Cho, J. Kim, and T. Hyun Kim, "Fast adaptation to super-resolution networks via meta-learning," 2020, *arXiv:2001.02905*.
[41] J. W. Soh, S. Cho, and N. I. Cho, "Meta-transfer learning for zero-shot super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3516–3525.
[42] X. Cheng, Z. Fu, and J. Yang, "Zero-shot image super-resolution with depth guided internal degradation learning," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 265–280.
[43] W. Zhang, Y. Liu, C. Dong, and Y. Qiao, "RankSRGAN: Generative adversarial networks with ranker for image super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3096–3105.
[44] X. He and Y. Peng, "Fine-grained image classification via combining vision and language," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5994–6002.
[45] K. Song, X.-S. Wei, X. Shu, R.-J. Song, and J. Lu, "Bi-modal progressive mask attention for fine-grained recognition," *IEEE Trans. Image Process.*, vol. 29, pp. 7006–7018, 2020.
[46] H. Xu, G. Qi, J. Li, M. Wang, K. Xu, and H. Gao, "Fine-grained image classification by visual-semantic embedding," in *Proc. 46th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 1043–1049.
[47] J. Noh, W. Bae, W. Lee, J. Seo, and G. Kim, "Better to follow, follow to be better: Towards precise supervision of feature super-resolution for small object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9725–9734.
[48] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
[49] R. Du *et al.*, "Fine-grained visual classification via progressive multi-granularity training of jigsaw patches," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 153–168.
[50] Z. Yang, T. Luo, D. Wang, Z. Hu, J. Gao, and L. Wang, "Learning to navigate for fine-grained classification," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 420–435.



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